AGRISMART AI – CROP SUITABILITY DOSSIER

Intelligent Crop Suitability & Soil Analysis Diagnostic System
Report Generated: September 27, 2026 - 20:51:44 UTC

RECOMMENDED CROP: RICE | **MODEL CONFIDENCE: 92.2%**

1. Field & Environmental Parameters Submitted

Parameter	Measured Value	Standard Unit	Agronomic Interpretation
Nitrogen (N)	90.0	kg/ha	High
Phosphorus (P)	42.0	kg/ha	Optimal
Potassium (K)	43.0	kg/ha	Optimal
Soil pH	6.50	pH (0-14)	Neutral
Temperature	20.9	°C	Thermal Regime
Relative Humidity	82.0	%	Atmospheric Moisture
Rainfall	202.9	mm	Tropical Wet / Humid

2. Top 5 Probabilistic Crop Candidates

Rank	Crop Candidate	Probability Score	Suitability Tier
#1	Rice	92.15%	Primary Choice
#2	Jute	7.85%	Secondary Option
#3	Apple	0.00%	Secondary Option
#4	Banana	0.00%	Alternative
#5	Blackgram	0.00%	Alternative

3. Soil Health & Fertility Diagnostic

NPK Stoichiometry: Total NPK load is **175.0 kg/ha** with an empirical ratio of **2.1 : 1.0 : 1.0**. Fertility Score: **90/100**.
Soil pH Status: Classified as **Neutral**. *Risk:* Optimal nutrient bioavailability for the majority of crops.
Amendment Advisory: Ideal soil reaction. Standard balanced fertilization.

4. Agro-Climatic & Abiotic Stress Assessment

Climate Classification: Tropical Wet / Humid. Microclimate Health Index: **100/100**.
Ecological Compatibility: Environmental regime aligns with the ecological niche of Rice.
Abiotic Management: Climatic conditions are stable with low abiotic risk.

Model Governance: Generated via AgriSmart AI Machine Learning Engine (Random Forest). Strictly validated on 2,200 non-fabricated empirical records. No data leakage.